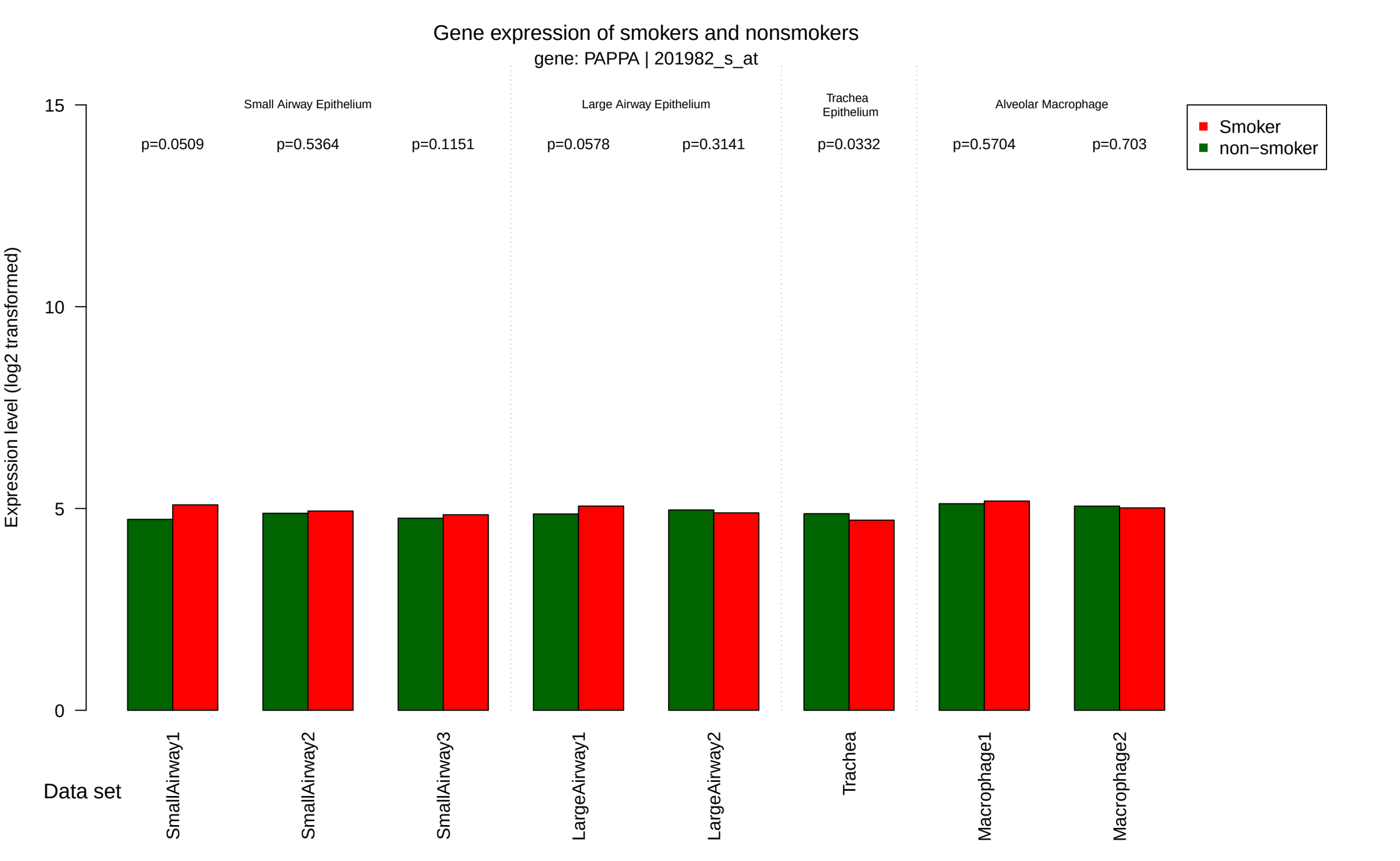
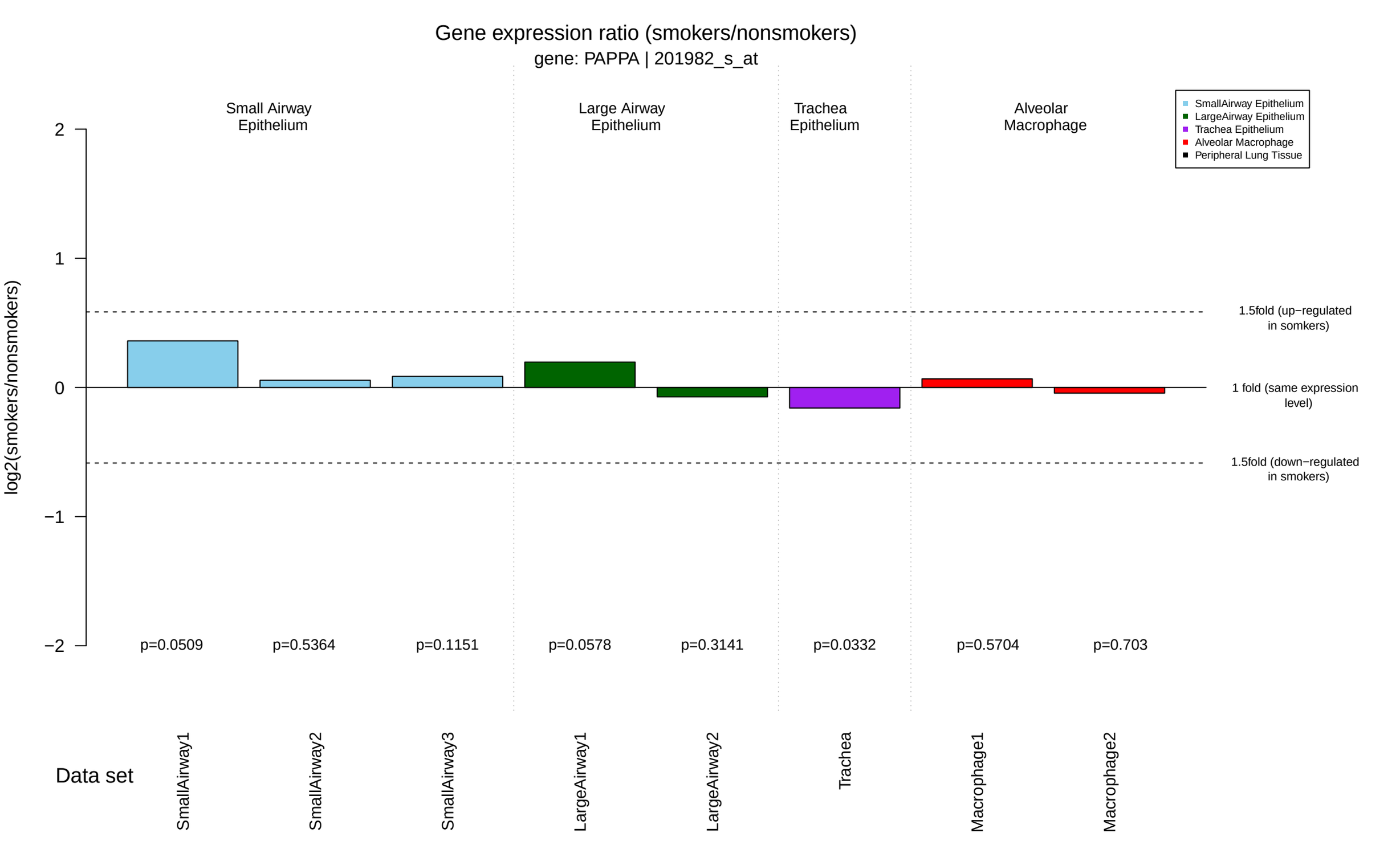
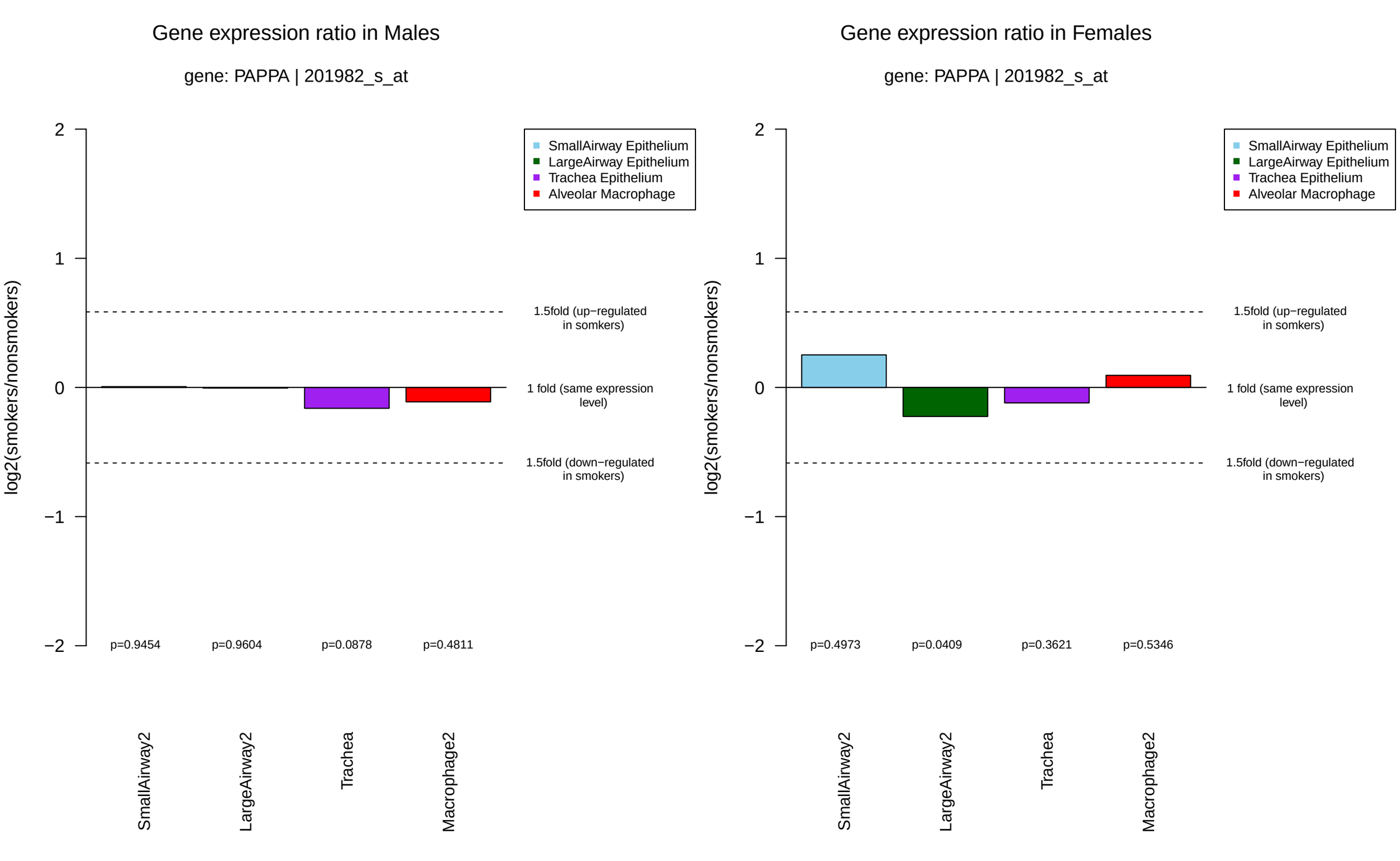
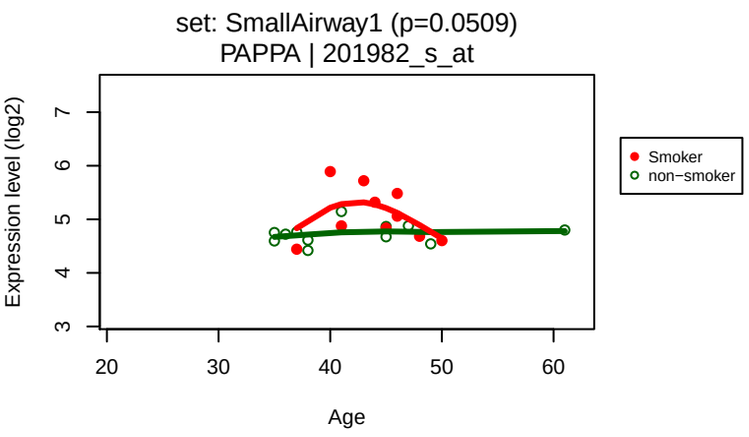
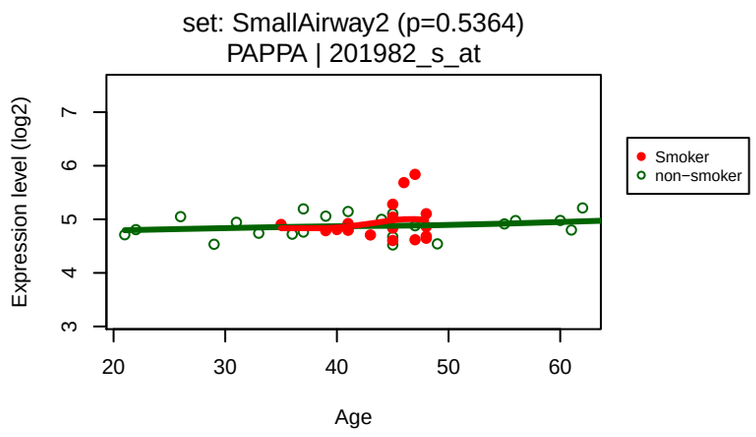
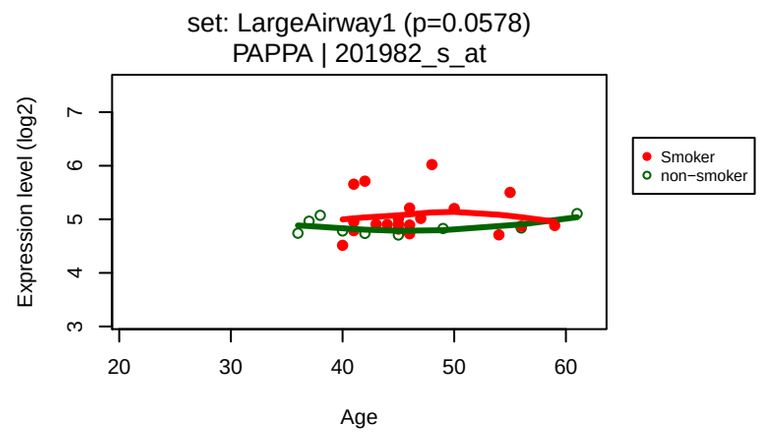

Gene expression of smokers and nonsmokers
gene: PAPPA | 201982_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0509
p=0.5364
p=0.1151
p=0.0578
p=0.3141
p=0.0332
p=0.5704
p=0.703
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

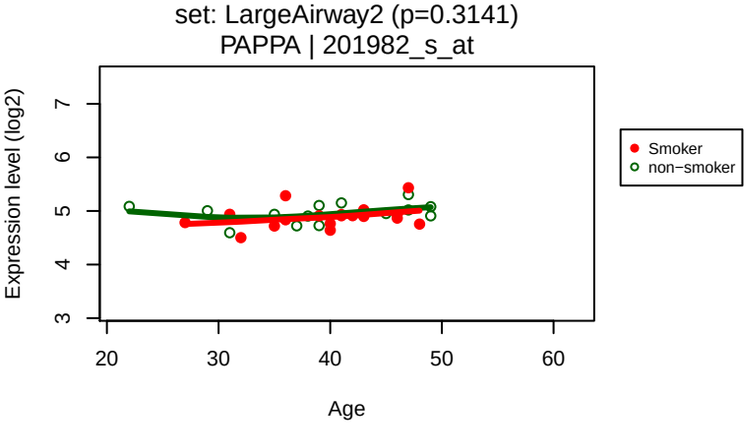

Gene expression ratio (smokers/nonsmokers)
gene: PAPPA | 201982_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0509
p=0.5364
p=0.1151
p=0.0578
p=0.3141
p=0.0332
p=0.5704
p=0.703
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

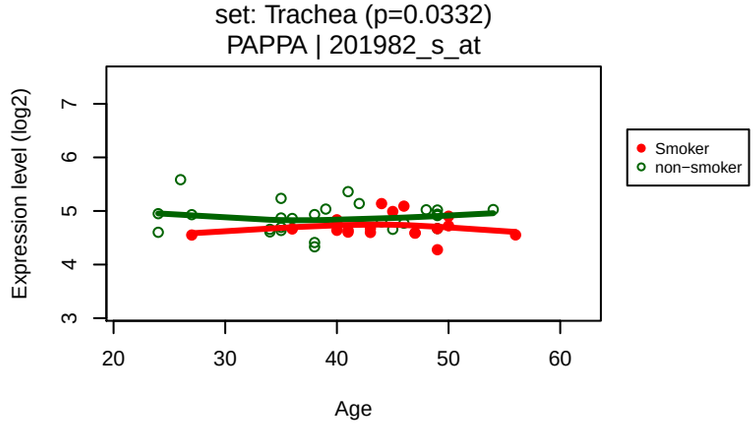

Gene expression ratio in Males
gene: PAPPA | 201982_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9454
p=0.9604
p=0.0878
p=0.4811
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAPPA | 201982_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4973
p=0.0409
p=0.3621
p=0.5346
SmallAirway2
LargeAirway2
Trachea
Macrophage2

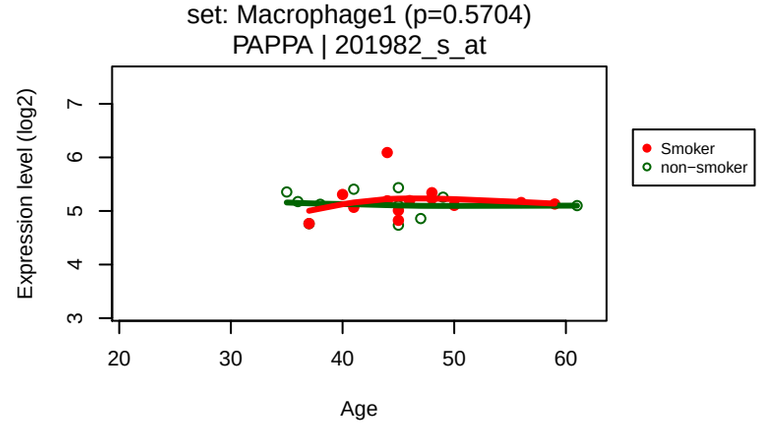

set: SmallAirway1 (p=0.0509)
PAPPA | 201982_s_at
Expression level (log2)
Age
Smoker
non−smoker

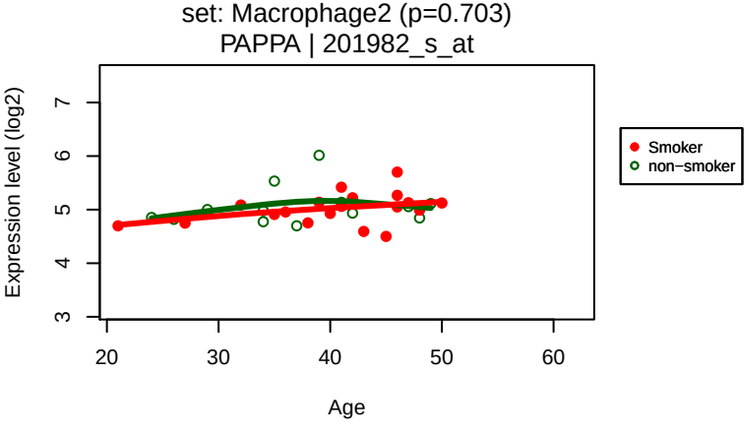

set: SmallAirway2 (p=0.5364)
PAPPA | 201982_s_at
Expression level (log2)
Age
Smoker
non−smoker

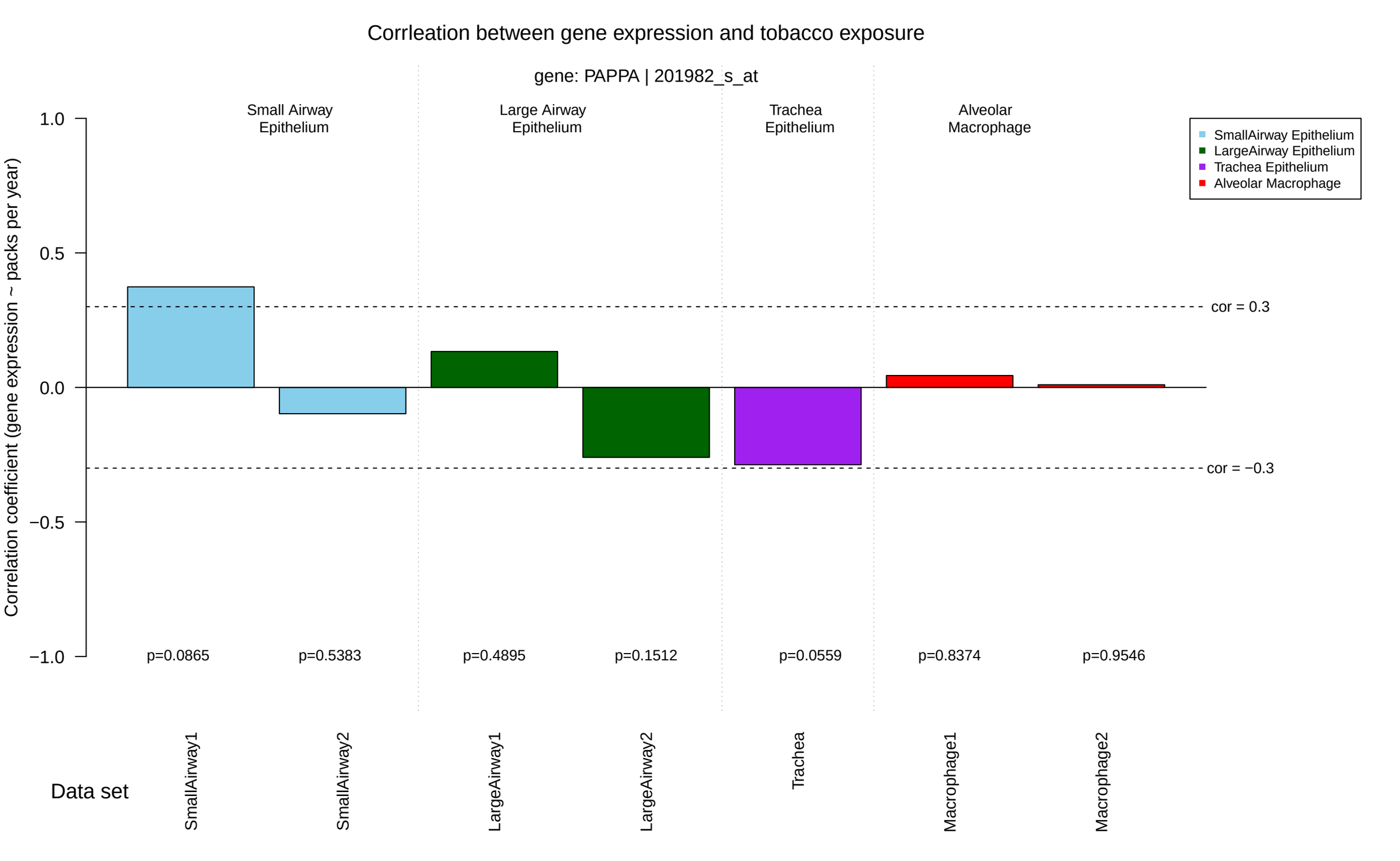

set: LargeAirway1 (p=0.0578)
PAPPA | 201982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3141)
PAPPA | 201982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0332)
PAPPA | 201982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5704)
PAPPA | 201982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.703)
PAPPA | 201982_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAPPA | 201982_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0865
p=0.5383
p=0.4895
p=0.1512
p=0.0559
p=0.8374
p=0.9546
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2